

The Modal Triplet Theory Program A1: Coherent Kinematics

Chart Persistence, Encoding Worldlines, and the Conditional Physical Bridge

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Abstract

We formulate coherent kinematics as an encoding-level construction and separate it from physical causal motion. An admissible chart is a typed reduction from an upper state domain to an effective description, with transition maps on controlled overlaps. Position is a declared localization functional of chart support. An encoding trajectory is generated by an actual upper path that remains representable through a compatible chain of charts; equivalent chart descriptions define the same encoding worldline. Regularity of this worldline requires regular upper evolution and regular chart and localization maps—contractivity alone is insufficient. Physical timelike and null classification, finite propagation, locality, and operator microcausality enter only after the corresponding Lorentzian, hyperbolic, or local-algebraic structures are supplied. Chart ordering alone is not a causal order. Effective mergers are failures of injectivity of an autonomous reduced map, while splits, selections, probabilities, and records require additional continuation, instrument, measure, and stability data. Failure of an exterior decoder or chart continuation does not imply termination of the underlying upper evolution or a physical worldline. These distinctions retain chart persistence as a useful pregeometric MTT language without deriving relativity, measurement outcomes, or a temporal arrow from projection alone.

Revision note for version 2

Supersedes. Program A1, version 1.0 (January 2026).

Reason. The first edition correctly identified chart persistence and worldline equivalence, but promoted descriptive continuation to physical causality, null and timelike structure, horizon formation, selection, records, and irreversibility without the required realization data.

Resolution. Version 2 types the upper path, chart transitions, localization maps, encoding worldlines, autonomous reduced maps, boundary continuation rules, and physical bridge separately.

Retained result. Compatible chart chains generated by an upper path define a rigorous encoding-level kinematics.

Remaining boundary. Physical trajectories, causal cones, microcausality, horizons, outcome selection, probabilities, records, and an arrow of time require their own selected laws and source hypotheses.

Contents

[Revision note for version 2](#)

1

1	Scope and Three Kinematic Levels	3
2	Admissible Charts and Transitions	3
3	Coherent Carriers, Support, and Position	4
4	Encoding Trajectories and Worldline Equivalence	4
5	Regularity Requires More Than Contraction	5
6	Conditional Bridge to Physical Spacetime Motion	5
7	Descriptive Order Is Not Causal Order	6
8	Effective Mergers, Splits, and Identity	6
9	Encoding Boundaries, Horizons, and Termination	6
10	Selection and Records Require Additional Data	7
11	Relation to the Corrected MTT Spine	7
12	Scoped Coherent-Kinematics Theorem	8
13	Conclusion	8

1 Scope and Three Kinematic Levels

Program A0 supplies a typed theory of reduced description: upper evolution, reduction, representative selection, exact decoding, autonomous descent, and effective merger are distinct constructions. The present paper asks a narrower question. Given compatible partial descriptions of an evolving upper state, what kinematic object is defined before a physical spacetime realization is chosen?

Three levels must remain separate.

Upper evolution. A path or flow in a declared MTT control or state space.

Encoding kinematics. Localization and persistence after applying compatible partial descriptions.

Physical kinematics. Curves, support propagation, and causal classes in a selected spacetime evolution law.

Encoding kinematics can represent physical motion after a bridge theorem. It does not itself provide a metric, null cone, velocity, force law, physical time, or local observable algebra. The stabilization parameter τ , a physical time t , an RG scale μ , and a shared-circle phase remain separately typed.

2 Admissible Charts and Transitions

Let X be an upper state space. An admissible chart is a triple

$$\mathfrak{C}_\alpha = (A_\alpha, Y_\alpha, r_\alpha), \quad r_\alpha : A_\alpha \longrightarrow Y_\alpha,$$

where $A_\alpha \subseteq X$ is a domain on which every declared operator, projection, stability, and interpretation margin is controlled. The map r_α is a reduction or encoding map; it need not be injective, linear, or a coordinate chart on X .

For $A_{\alpha\beta} = A_\alpha \cap A_\beta$, an exact transition map is a map

$$h_{\beta\alpha} : r_\alpha(A_{\alpha\beta}) \longrightarrow r_\beta(A_{\alpha\beta})$$

satisfying

$$r_\beta(x) = h_{\beta\alpha}(r_\alpha(x)), \quad x \in A_{\alpha\beta}.$$

It exists exactly when r_β is constant on the r_α fibers inside the overlap. This is the projection-descent criterion applied to two descriptions. Approximate compatibility must instead state a metric, an error bound, and an accumulation rule.

Assumption 2.1 (Compatible chart family). On every nonempty overlap, the required transition maps exist, are continuous in the declared topologies, are mutually inverse on their attained overlap images, obey $h_{\alpha\alpha} = \text{id}$, and satisfy

$$h_{\gamma\beta}h_{\beta\alpha} = h_{\gamma\alpha}$$

where all terms are defined.

Under Assumption 2.1, the disjoint union of the attained chart images may be quotiented by the equivalence relation generated by the transitions. Denote this effective descriptive space by $\mathfrak{Y}$. This constructs an atlas of descriptions. It does not turn X or $\mathfrak{Y}$ into physical spacetime.

3 Coherent Carriers, Support, and Position

A coherent carrier $\mathcal{B}$ is a declared upper subset or labeled family controlled by the hypotheses of the application. It may be an invariant basin, a localized solution family, or a neighborhood of a selected equilibrium. These cases are not identified without an additional theorem.

Its attained support in chart α is

$$\text{Supp}_\alpha(\mathcal{B}) := r_\alpha(\mathcal{B} \cap A_\alpha).$$

Because r_α may be noninjective, equal supports do not prove equality of upper carriers. Conversely, different chart supports may describe the same carrier through a transition map. No exclusion principle follows from support coincidence alone.

Definition 3.1 (Chart-relative position). Let $\mathcal{S}_\alpha$ be a declared class of localized support sets in Y_α . A position diagnostic is a map

$$\ell_\alpha : \mathcal{S}_\alpha \longrightarrow Z_\alpha$$

with specified domain, covariance, and regularity under chart transitions. The chart-relative position of $\mathcal{B}$ is

$$\ell_\alpha(\text{Supp}_\alpha(\mathcal{B}))$$

when the support lies in $\mathcal{S}_\alpha$.

A chart-independent position exists only if the target spaces and their transition maps are supplied and the diagnostics agree on every attained overlap. Thus position is naturally an equivalence class of admissible chart representations, not an automatically global coordinate.

Centroids require further geometry. On a Riemannian base they should be intrinsic Karcher means inside a convex normal ball, as in Fixed Points IV, rather than unqualified coordinate averages.

4 Encoding Trajectories and Worldline Equivalence

Let J be an interval of an explicitly named parameter s , and let $x : J \rightarrow X$ be an upper path. This path may be a stabilization trajectory, a physical solution lifted to the upper space, or another declared family.

Definition 4.1 (Chart-persistent encoding trajectory). The path x has a chart-persistent encoding on J if there is an interval cover $\{J_i\}$ and chart labels α_i such that $x(J_i) \subseteq A_{\alpha_i}$ and

$$z_i(s) = r_{\alpha_i}(x(s))$$

satisfies

$$z_j(s) = h_{\alpha_j \alpha_i}(z_i(s))$$

on every interval and chart overlap.

Definition 4.2 (Encoding worldline equivalence). Two chart-persistent families represent the same encoding worldline if their curves agree through the declared transition maps on a common refinement of their interval covers. A monotone reparameterization may also be quotiented out when the application declares parameter-free curve equivalence.

Theorem 4.3 (Gluing of encoding trajectories). *Under Assumption 2.1, a chart-persistent family defines a unique curve in $\mathfrak{Y}$ up to encoding-worldline equivalence. Conversely, a curve in $\mathfrak{Y}$ has local compatible chart representatives, but those representatives need not determine a unique upper path.*

Proof. The transition and cocycle equations identify the chart representatives on double and triple overlaps, so $s \mapsto [z_i(s)]$ is independent of the active chart. Local representatives give the converse. Noninjectivity of the reductions prevents unique upper recovery in general. $\square$

The underlying upper path is essential. An arbitrary combinatorial chain of nonempty chart overlaps records descriptive adjacency; it does not by itself describe motion of any state.

5 Regularity Requires More Than Contraction

Theorem 5.1 (Regular encoding trajectory). *Suppose the relevant spaces carry declared metrics, $x : J \rightarrow X$ is locally Lipschitz, each active r_α is Lipschitz on the attained path, each transition map is Lipschitz on the attained overlap, and each localization diagnostic ℓ_α is Lipschitz with respect to a declared support metric. Then every chart and position representative of the encoding worldline is locally Lipschitz, with constants bounded by the corresponding composition products.*

Proof. Compose the Lipschitz upper path with the chart and localization maps. On an overlap, compose once more with the Lipschitz transition map. $\square$

A contraction estimate for a time-step map controls separation of iterates in the chosen state norm. By itself it proves neither continuity of the parameter path nor continuity of a chart, support map, or localization functional. Those regularity hypotheses cannot be omitted.

6 Conditional Bridge to Physical Spacetime Motion

Adopt a physical realization only in this section: a globally hyperbolic Lorentzian base (Y_4, g) and a selected local physical equation with well-posed Cauchy evolution. Its gauge-fixed principal symbol determines its characteristic cone.

Definition 6.1 (Physical causal class). Using the mostly-plus convention, a differentiable physical curve $\gamma : I \rightarrow Y_4$ is timelike, null, or spacelike when $g(\dot{\gamma}, \dot{\gamma})$ is respectively negative, zero, or positive. This definition belongs to the selected Lorentzian realization, not to chart overlap alone.

Theorem 6.2 (Conditional finite propagation). *Suppose the selected physical equation has a finite-domain-of-dependence theorem with causal cone determined by g , and let a localized solution have initial support K on a Cauchy surface. Its future support is contained in $J^+(K)$ and its past support in $J^-(K)$ on the corresponding evolution domains. A coherent reduction preserves this conclusion only when the principal-symbol descent and locality hypotheses are also proved.*

This is the available no-superluminal-support statement [7]. It is conditional on the physical PDE and concerns support propagation. It is not proved by admissibility, contractivity, or a chart partial order.

Operator microcausality is a different statement: it concerns commutators of local observables at spacelike separation. It requires an appropriate local classical or quantum observable net and a theorem that the reduction preserves its locality. Finite support propagation alone does not establish it.

A localization diagnostic may represent a physical worldline only when existence, uniqueness, and regularity of that localization are proved. Wave packets and extended supports need not have a unique point trajectory.

7 Descriptive Order Is Not Causal Order

One may partially order charts by domain inclusion, refinement, or allowed continuation. Such an order records descriptive availability. A physical causal order records reachability by causal curves or support propagation in a Lorentzian or hyperbolic realization. Neither order determines the other without a monotone bridge theorem.

Likewise, an oriented stabilization flow can provide an ordering in τ but not a physical arrow of time. A temporal arrow requires a physical evolution plus an asymmetric law or boundary condition, such as a Lyapunov or entropy theorem in its proper domain.

A compact shared $U(1)$ circle may synchronize phase or holonomy across modal lanes. Its angular coordinate is not physical time. Relating it to temporal ordering requires a declared lift $\mathbb{R} \rightarrow S^1$, a selected noncompact evolution parameter, and a theorem showing how the lifted order enters the physical propagator.

8 Effective Mergers, Splits, and Identity

Suppose autonomous descent gives a reduced map $F_s : Y_0 \rightarrow Y_s$.

Definition 8.1 (Effective merger). An effective merger occurs when $y_1 \neq y_2$ but $F_s(y_1) = F_s(y_2)$. The final effective state does not uniquely decode the prior effective label.

This is a statement about the reduced map. It does not prove physical coalescence, attractive interaction, destruction of either upper state, entropy production, or equality of the corresponding upper states.

A deterministic single-valued map sends one state to one output and cannot by itself encode a split into alternative outcomes. A split description requires a refined state space, a multivalued relation, a stochastic kernel with a measure, or a hybrid branch/reset law. Branch probabilities do not follow from the number of chart continuations.

Identity across charts is therefore a declared equivalence supported by transition data and diagnostics. If the diagnostic family is not separating, indistinguishability does not prove upper identity.

9 Encoding Boundaries, Horizons, and Termination

For a fixed upper path, a chosen compatible chart family may have a maximal representation interval $J_{\text{enc}} \subseteq J$. An endpoint of J_{enc} is an *encoding boundary*: beyond it, that family no longer supplies a compatible representation. This is a precise replacement for the first edition’s generic terminal-worldline language.

In a physical spacetime, an event horizon does not generally terminate an infalling timelike worldline at horizon crossing [8]. What may end is its representation in a chosen exterior chart

or the ability of exterior data to decode interior information. Singular endpoints, geodesic incompleteness, chart boundaries, and failures of the physical PDE are separate notions.

Thus horizon language in MTT must specify whether the failure is:

1. loss of a compatible chart continuation;
2. loss or ill-conditioning of a representative extension;
3. failure of an exact exterior decoder;
4. breakdown of the physical evolution law; or
5. genuine physical geodesic incompleteness.

Noninjectivity of an exterior reduction obstructs exact decoding on any nontrivial fiber, but it does not forbid a representative section. No global-right-inverse obstruction follows from noninjectivity alone.

An encoding boundary therefore terminates one representation contract. It does not imply termination of the upper path, physical annihilation of its carrier, or an irreversible physical event. A “nil” label may name the failed encoding state only after its diagnostic has been explicitly defined.

10 Selection and Records Require Additional Data

Crossing an admissibility boundary detects failure of a declared chart contract. It does not choose what happens next. A continuation or reset rule must be supplied separately. If $m_\alpha : \overline{A_\alpha} \rightarrow \mathbb{R}$ is the named continuous admissibility margin, write

$$\partial_{\text{adm}} A_\alpha := \{x \in \overline{A_\alpha} : m_\alpha(x) = 0\}$$

for its declared exit set. A continuation rule may then be typed, for example, as

$$\mathcal{R} : \partial_{\text{adm}} A_\alpha \rightrightarrows \coprod_{\beta} A_\beta,$$

with a single-valued map, a multivalued relation, a deterministic hybrid law, or a Markov kernel according to the intended application.

If probabilities are required, the initial measure and transition kernel must be stated. Neither fiber cardinality nor basin count fixes outcome weights. If one realized continuation is to be called an outcome, a selection instrument must identify the relevant observable event.

A record additionally requires a memory variable or observable whose distinct values remain stable for a declared duration under the selected evolution. Chart persistence can represent such a record after this stability theorem is proved; it does not create records merely by losing an inverse or crossing a margin.

11 Relation to the Corrected MTT Spine

The corrected MTT papers divide the work as follows.

- Program A0 and the Projection–Admissibility paper type reduction, descent, representative sections, exact decoders, and effective merger.
- The Foundation supplies typed state spaces, coherent projection, separate evolution parameters, and explicit admissibility margins.

- Fixed Points I–VI supply conditional fixed points, equilibrium promotion, damping, leakage, intrinsic centroids, and frozen covariance.
- Signature Stability supplies the conditional physical principal-symbol bridge.

Consequently, chart persistence is a valid encoding-level construction. Physical locality and null/timelike classes come from the selected hyperbolic realization. Effective merger does not replace the signed-interaction and dynamical-connection requirements of Fixed Points IV. An admissibility exit does not select a successor chart or physical state.

Quantum measurement, photons, QFT propagation, black-hole exteriors, and cosmological horizons may instantiate this framework only after their own state, evolution, observable, and source maps have been constructed. They are not derived here.

12 Scoped Coherent-Kinematics Theorem

Theorem 12.1 (Encoding kinematics with a physical bridge). *Assume a compatible admissible chart family and a locally Lipschitz upper path satisfying Theorem 5.1. Then the path defines a regular encoding worldline up to chart-transition equivalence. If, additionally, the upper path is a lift of a localized solution of a selected globally hyperbolic physical equation and the coherent principal-symbol and locality descent hypotheses hold, the encoding represents physical support propagation within the supplied causal cone.*

Proof. The first conclusion follows from Theorems 4.3 and 5.1. The second follows from Theorem 6.2 and the assumed intertwining of the physical principal symbol, support, and reduction maps. $\square$

The first conclusion is encoding kinematics. The second is a conditional physical bridge. Neither conclusion derives the metric, physical equation, particle identity, unique centroid, merger outcome, selection probability, record, horizon, microcausal operator algebra, or temporal arrow.

13 Conclusion

Chart-relative position and persistence remain useful MTT concepts once their status is kept precise. They describe how an upper carrier is represented through compatible partial encodings. Physical motion requires more: a local hyperbolic equation, a causal principal symbol, finite propagation, and a controlled localization map. Selection and records require instruments, measures, and stable memory observables. With these distinctions, worldline, horizon, merger, and split language can be used without confusing loss of description with physical termination or outcome selection.

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